INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

SCHEME OF INSTRUCTION

Diploma in Engineering First Year

I - YEAR, I-SEMESTER (Effective from the academic session 2017-2018)

S. No.	Course Code	SUBJECT	PERIODS			Evaluation Scheme					Total	Credit
			L	T	P	CT	TA	AT	Total	E-Sem		
	THEORY											
1	DAS101	Mathematics-I	4	0	0	20	10	10	40	60	100	4
2	DAS102 OR DAS103	Physics OR Chemistry	6	0	0	20	10	10	40	60	100	4
3	DEE101 OR DMC101	Fundamentals of Electrical & Electronics Engineering OR Fundamentals of Mechanical& Civil Engineering	4	0	0	20	10	10	40	60	100	4
4	DCS101 OR DME101	Fundamentals of Computer OR Fundamentals of Mechanics	4	0	0	20	10	10	40	60	100	4
	PRACTICAL/TRAINING/PROJECT											
5	DAS152 OR DAS153	Physics Lab OR Chemistry Lab	0	0	3	-	-		50	50	100	2
6	DEE151 OR DMC151	Electrical & Electronics Lab OR Civil & Mechanical Lab	0	0	2	-	-		50	50	100	2
7	DCS151 OR DME151	Computer Concept Lab OR Workshop Practice	0	0	2	-	-		50	50	100	2
8	DCE151	Engineering Drawing	0	0	2	-	-		50	50	100	2
9	GP101	GeneralProficiency	-	-	-	-	-		-	50	50	1
		Total	**18**	**0**	**7**				**360**	**490**	**850**	**25**

L-Lecture, **T**-Tutorial, **P**- Practical, **CT**-Cumulative Test, **TA**- Teacher Assessment, **AT-Attendance, E-Sem**- End Semester Marks

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SCHEME OF INSTRUCTION

SCHEME OF INSTRUCTION

Diploma inEngineering First Year

Diploma in Engineering First Year

I - YEAR, II-SEMESTER (Effective from the academic session 2017-2018)

S. No.	Course Code	SUBJECT	PERIODS			Evaluation Scheme					Total	Credit
			L	T	P	CT	TA	AT	Total	E-Sem		
	THEORY											
1	**DAS201**	**Mathematics-II**	4	0	0	20	10	10	40	60	100	4
2	**DAS203 OR DAS202**	**Chemistry OR Physics**	6	0	0	20	10	10	40	60	100	4
3	**DMC201 OR DEE201**	**Fundamentals of Mechanical& Civil Engineering OR Fundamentals of Electrical & Electronics Engineering**	4	0	0	20	10	10	40	60	100	4
4	**DME201 OR DCS201**	**Fundamentals of Mechanics OR Fundamentals of Computer**	4	0	0	20	10	10	40	60	100	4
	PRACTICAL/TRAINING/PROJECT											
5	**DAS253 OR DAS252**	**Chemistry Lab OR Physics Lab**	0	0	3	-	-		50	50	100	2
6	**DMC251 OR DEE251**	**Civil & Mechanical Lab OR Electrical & Electronics Lab**	0	0	2	-	-		50	50	100	2
7	**DME251 OR DCS251**	**Workshop Practice OR Computer Concept Lab**	0	0	2	-	-		50	50	100	2
8	**DCE251**	**Engineering Drawing**	0	0	2	-	-		50	50	100	2
9	**GP201**	**GeneralProficiency**	-	-	-	-	-		-	50	50	1
		Total	**18**	**0**	**7**				**360**	**490**	**850**	**25**

L-Lecture, **T**-Tutorial, **P**- Practical, **CT**-Cumulative Test, **TA**- Teacher Assessment, **AT-Attendance, E-Sem**- End Semester Marks

LTP
310

DCS-101/201 Computer Fundamentals

Unit-I

Introductionto Computer
CharacteristicsofComputers,WhatComputerscando,Whatcomputerscan'tdo,Classificationof DigitalComputerSystems,AnatomyofaDigitalComputerUsageofcomputersystemindifferent domainslikeoffice,bookpublication,ticketreservation,banksetc.ComponentsofPC–Mouse, keyboard, CPU, monitor,printers, scanners, modem, memory,sound cards, pen drives.

Unit-II
Introduction to OperatingSystem(Windows 7)
WorkingwithWindowsdesktop,icons,taskbar,menubaroptions,MyDocuments,MyComputer, Control Panel, Recycle bin Concept of drives, folders, files Windows accessories –Notepad, WordPad, paint, clock,calendar,calculator

Unit-III
Data Representation- Representing different symbols, minimizing errors, Representing more Symbols,GenericFormula,theASCIIcode,theEBCDICcode,RulesofDecimalnumberSystem anditsconversiontoBinary **Multimedia-**Digitalimages,analogtodigitalconversions,digital audioand digital video

Unit-IV

GUI BasedSoftware –MS– Office 2010
MS-Word–Openingmenus,toolbars,openingandclosingdocuments,clipboardconceptMS–Excel– Workingandmanipulatingdatawithexcel,formulas,functions,chartanditstypesMS–PowerPoint– WorkingwithPowerPointandpresentation,Changinglayout,Graphs,Autocontent wizard ,Slideshow, Animationeffects, Normal,outline, Slide sorter, Readingview.

Unit-V
Internet
History ofInternet,equipmentsrequiredforInternetconnection,browser(InternetExplorer,Mozilla and Firefox, Google Chrome)**Algorithms**-Introduction, Three Basic Operations, Proceduresand Programs
VDUandPrinters-Human-computerinterface,Keyboard,RasterScanning,FrameBuffer,Basicsof Graphics,BlackandWhite/ColorTerminals,Textbasedterminals,LEDs/LCDs,InkjetPrinters, Laser Printer

Learning Resources:
1. Books:

No	Author	Title	Publisher
1	AchyutGodbole	DemystifyingComputer	TMH
2	AlexisLeon	Introductionto Computers	Vikas PublishingHouse
3	Vikas Gupta	Comdex Computer Course Kit	DreamtechPress
4	Steve Schwartz	Microsoft Office 2010	Pearson
5	Elaine Marmel	Microsoft Project 2010 (Bible)	Wiley India
6	Preppernau Cox	Windows7Step byStep	PHI

DEE-101/201Fundamentals of Electrical & Electronics Engg.

LTP
4 0 0

Unit 1

Basic Terminologyand their concepts: Current, EMF, potential difference (voltage), resistance, resistivitytheirunitsconductors& insulators,Insulationresistanceofacable.Effectoftemperatureonthe resistanceofconductors,temperaturecoefficientofresistance.Electricalpower,energyandtheirunits (SI),Heatingeffectofelectriccurrentanditspracticalexamples.Relationshipbetween electrical,mechanical andthermal SIunits of work, power andenergy,Electrical Safetyand precautions.

Unit 2

D.C. Circuits: Kirchhoff'slaws.Differenttypesofvoltageandcurrentsources,Introductionto Thevenin, super-position, and maximum power transfer theorem.

Unit 3

A.C. Circuits: Instantaneousvalue,maximum(peak)value,cycle,frequency,alternatecurrentand voltage.Equationofanalternatingvoltageandcurrentandwaveshapevaryingsinusoidally.Averageand RMSvalueofalternatingvoltageandcurrent.Conceptofphase,phasedifferenceandphasor representation of alternating voltageand current with pure resistance, inductance, capacitance.

Unit4

Components andDiodes : Basic concepts of energy bands in materials, concept of forbidden gap ,Intrinsicand extrinsicsemiconductors, donors and acceptors impurities
Junctiondiode:p-njunction,depletionlayer,v-icharacteristics,dioderesistance,capacitancediode ratings.

Unit5

Breakdowndiodes:breakdownmechanisms(zenerandavalanche) breakdowncharacteristics,zener resistance, zener dioderatings
DiodeApplications:rectifiers(halfwaveandfullwave),calculationoftransformerutilizationfactorand diode ratings.

Unit6

Bipolar JunctionTransistor
Basicconstruction,transistoraction CB,CE andCCconfigurations,input/outputcharacteristics,basicsof JFET and MOSFET.
Switching theory and logic design
Number systems, conversion ofbasesBoolean algebra,logicgates, concept of universal gate.

ReferenceBooks:

1. obertL.Boylestad/LouisNashelsky"ElectronicDevicesandCircuitTheory",9thEdition, Pearson Education 2007
2. DevidA.Bell"ElectronicDevicesandCircuits",5thEdition,OXFORDUniversityPress2008
3. MorrisMano"DigitalComputerDesign",PHI2003
4. A Text Book ofElectrical technology: B.L.Theraja, A.K.Theraja.,Volume-I, S.Chand Publication.
5. asic Electricaland Electronics Engineering: ByG. Rajalakshmi, N. Srivanandham.
6. Fundamentals of Electrical and Electronics Engineering: BySmarajitGhosh.
7. Basic Electricaland Electronics Engineering: byJ.B. Gupta.

ENGINEERINGDRAWING/DCE151

NOTE: LatestIndian StandardsCode of Practice to be followed.

1.0 Drawing, instruments andtheiruses.1.1 Introductionto various drawing, instruments.
1.2 Correctuseand careof Instruments 1.3 Sizes ofdrawingsheets and theirlayouts.

1. **(a) Lettering Techniques1 Sheet**
   Printingofverticalandinclined,normalsinglestrokecapitalletters.Printingof Verticaland inclinednormalsingle stroke numbers. Stencilsand their use.
   (b) Introduction to Scales1 Sheet
   Necessityanduse,RFTypesofscalesusedingeneralengineeringdrawing. Plane,diagonaland chord scales.
2. **ConventionalPresentation : 2 Sheet**
   Thread(InternalandExternal),Weldedjoint,Typesoflines,Conventional representation ofmaterials,Conventionalrepresentation of machineparts.
3. **(a) Principles of Projection 1 Sheet**
   Orthographic,Pictorialandperspective.Conceptofhorizontalandvertical planes.DifferencebetweenIandIIIangle projections.Dimensoningtechniques.
   (b) Projections of points, lines andplanes. 1 Sheet
4. **(a)OrthographicProjections of Simple :3 Sheet**
   GeometricalSolidsEdgeandaxismakinggivenangleswith the referenceplanes. Facemakinggivenangleswithreferenceplanes.Faceanditsedgemakinggiven angles with referanceplanes.
   (b) Orthographic viewsof simplecompositesolids fromtheir isometricviews.
   (c) Exercises onmissing surfaces andviews
5. **Section of Solids:1 Sheet**
   ConceptofsectioningCasesinvolvingcuttingplaneparalleltooneofthe referenceplanesandprependiculartotheothers.Casesinvolvingcuttingplane perpendiculartooneofthereferenceplanesandinclindtotheothersplane,true shapeofthe section.
6. **Isometric Projection: 2 Sheet**
   Isometric scale Isometric projection ofsolids.
7. **Free handsketching: 1 Sheet**
   Use ofsquared paper Orthographic viewsof simple solids Isometric views of simplejob like carpentaryjoints
8. **Development of Surfaces:1 Sheet**
   Parallellineandradiallinemethodsofdevelopments.Developmentofsimple and truncated surfaces(Cube, prism, cylinder, cone and pyramid).
9. **Assembly andDisassembly Drawings: 2 Sheet**
   Plummerblock,FootstepbearingsCouplingsetc.Riveted&WeldedJoints Screw and formof screw thread
10. **Orthographic ProjectionOf Machine Parts: 2 Sheet**
    Nutand Bolt,Lockingdevice, Wallbracket
11. **PractiveOn Auto Cad :**
    Todrawgeometricalfiguresusingline,circle,arc,polygon,ellipse,rectangle-eraseandothereditingCommands and osnapcommands (two dimensional drawingonly)

    NOTE

    Thedrawiangshouldincludedimensionwithtolerancewhereevernecessary, materiallistaccordingtoI.S.code.25%ofthedrawingsheetshouldbedrawnin firstangle projectionandrest75% drawingsheetshould be inthirdanglefigure.

DIPLOMAINENGINEERING
APPLIED PHYSICS(DAS-102/202)

LTPC
4 004

(First/Second Semester)

CourseContents:

Unit-I

Vector:Scalarandvectorquantities:Addition,Subtraction,AndResolutionofvector-Cartesiancomponentsofvector,Scalarandvectorproductoftwovectors.
ForceandMotion:Parabolicmotion,projectilesthrownhorizontallyandatanangle. Problemsontimeofflight,horizontalrangeandverticalheight.Circularmotion,angular velocity,angularaccelerationandcentripetalacceleration.Relationshipbetweenlinearand angularvelocityandacceleration.Centripetalandcentrifugalforces.Gravitationalforce, Kepler'slaws,Escapevelocity,geostationary satellite.

Unit-II

DynamicsofRigidBody(RotationalMotion):Rotationalmotion,Momentofinertia, Theorems(PerpendicularandParallelaxis)ofmomentofinertia(Statement),Radiusof gyration,angularmomentum,Conservation ofangularmomentum,Torque,Rotational kineticenergy.
Friction:Introduction,Advantageanddisadvantageoffriction,Staticanddynamicfrictional forces.
FluidMechanics:Surfacetension,Equationofcontinuity($A_1V_1=A_2V_2$),Bernoulli's theorem,streamlineandTurbulentflow,Viscosity,coefficientofviscosity&its determinationbyStock'smethod.

Unit-III

Elasticity:-Elasticity,stressandstrain,Hook'slaw,elasticlimit,Yieldingpointand breakingpoint.Modulusofelasticity:Young'smodulus,bulkmodulusandmodulusof rigidity,Poissonratio.
SimpleHarmonic Motion: PeriodicMotion, characteristicsofsimple harmonic motion; equation ofS.H.M.anddetermination ofvelocityandacceleration.Simplependulum. DerivationoftheirperiodicTime.KineticEnergyandPotentialEnergyinS.H.M.Energy conservationinS.H.M.

Unit-IV

ApplicationofSoundWaves:-Acoustics:Standingwaves,ClosedandOpenorganpipes, Resonance.Echoandreverberationandreverberationtime.Sabine'sformula.Controlof reverberationtime.
Optics:Quantumnatureoflight,Coherence,Dualityofwaveandparticle,Concept of Interference,Fraunhoffersingle-slitdiffraction,Elementaryconceptofpolarization.

Unit-V

Electrostatics:ElectricCharges,Conservationlawofcharge,Coulomb'slaw-forcebetweentwopoint charges,superpositionprincipleandcontinuouschargedistribution. Electricfield,electricfieldduetoapointcharge,electricfieldlines,electricdipole. Electricflux,statementofGauss'stheorem.Electricpotential,potentialdifference,electric potentialduetoapointcharge,equipotential surfaces.
Electrodynamics:Electromotiveforce,Ohm'slaw,LimitationsofOhm'slaw,Ampere's Law,faraday's law,Biot-Savart'sLaw.

TextBooks:

1. KumarTyagi,AppliedPhysics,NavbhartiPrakashan,Meerut.
2. KushwahaP.S,AppliedPhysics,BharatBhartiPublication,Meerut.
3. JainVibhaAppliedPhysics,DhanpatRai&Company(P)Ltd.,Delhi.

ReferenceBooks:

1. GaurR.K&GuptaS.L.,EngineeringPhysics,DhanpatRaiPub.,NewDelhi
2. GaurR.K.BasicAppliedPhysics,DhanpatRaiPub.,NewDelhi

Note:This syllabus is designedaccordingly6lectures perweek.

MATHEMATICS–I

FirstSemester

	L	T	P	C
CourseCode:DAS101	4	-	-	4

UnitI

SERIES:A.P.andG.P.;n^{th}term,Sumtonterms,ArithmeticMean.Binomialtheoremforpositive, negativeandfractionalindex(withoutproof).ApplicationofBinomialtheorem.

UnitII

DETERMINANTS:-Elementarypropertiesofdeterminantsoforder2and3,multiplicationsystemof algebraicequations,Consistencyofequation,Crammer'srule.

VECTORALGEBRA:-DotandCrossproductoftwovectors,Scalarandvectortripleproducts.Workdone, Momentofaforce.

UnitIII

TRIGONOMETRY:-

Relationsbetweensidesandanglesofatriangle:Statementofvariousformulaeshowingrelationshipbetweensidesan danglesofatriangle.Complexnumbers,Representation,Modulus andamplitudeDeMoivre'stheorem,itsapplicationinsolvingalgebraicequations,ModulusFunctionanditsprope rties.

UnitIV

CO-ORDINATEGEOMETRY-I:StandardformofCircle,Parabola,EllipseandHyperbola. Tangentandnormaltothesecurves.

UnitV

CO-ORDINATEGEOMETRY-II Straightlines,planesandspheresin3–dimensionalspace.Distancebetween twopoints inspace, directioncosinesanddirection ratios,projectionsfindingequationofa straightline,andshortestdistancebetweentwolines.

Differentforms ofplanes represented byequation lx+ my+ nz=c, relation between linesandplanes, sphere $x^2 + y^2 + z^2 + 2gx + 2fy + 2wz=d$.

TextBooks:

1. Sharma,RD,*AppliedMathematics*.
2. GrewalBS,*Elementary EngineeringMathematics*,KhannaPublication.

ReferenceBooks:

1. GorakhPrasad,*Differential&IntegralCalculus*
2. MittalSC&Mittal,SK.,*TwoDimensionalCoordinate*,PragatiPrakashan,Meerut
3. Loney,S L,*Trigonometry(Ipart)*
4. Goel,BS,*Algebra*

APPLIED CHEMISTRY

First/Second Semester **L TP C**

CourseCode:DAS103/203 **4 - - 4**

CourseContents:

Unit:-I

ATOMIC STRUCTURE

Basicconceptofatomicstructure,Matterwaveconcept,deBrogliewaveequation,Quantum numbers,Heisenberg'sUncertainty Principle,Shapesoforbitals.

CHEMICALBONDING:-Overviewofbasicconcept,Ionic,Co-valentandCo-oridination Bond,MolecularOrbitalTheoryanditsapplicationstoHomoandHeterodiatomicmolecules,Hydrogenb ondinganditsapplications,ValenceBondTheory,HybridisationandGeometrical shapeof $BeCl_2$, NH_3, CH_4molecules.

(16Lectures)

Unit:-II

ELECTRO CHEMISTRY: Arrhenius's Theoryof electrolytic dissociation, Electrolytic conductance,Oswalddilutionlaw.ConceptofAcidandbases:Bronsted,Arrhenius'sandLewistheory. ConceptofpHanditmeasurementbypHmeter.Buffersolutions,Indicators,Solubility product,Redoxreactions,Electrodepotential(NernstEquation),Electro-chemicalcell(Galvanic and Electrolytic). Standard electrode potential, Electro chemical series andits application.

(11Lectures)

Unit:-III

ENVIRONMENTALPOLLUTIONANDITSCONTROL: Conceptandvarioustypesof environmentalpollutionwithspecialreferencetoairpollutionandwaterpollution.General measurestocontrolenvironmentalpollution.depletionofOzonelayer,Greenhouseeffect,Acid rain, Smog formation, Chemical and photochemical reaction, Various species in atmosphere.

(9 Lectures)

Unit:-IV

WATERTREATMENT: Conceptofhardandsoftwater,Hardnessofwater,Itslimitsand determinationofhardnessofwaterbyEDTAmethod.Softeningmethods(ZeoliteandIon exchangeresinprocess).Disadvantageofhardwaterindifferentindustries,Boilerfeedwater boilerscaleformation,Corrosion,Causticembritilment,primmingandfoaming.Characteristics imparted byvarious impurities or contaminants such as colour,
odour, taste and sediments andtheir analysis.

CORROSION: Conceptofmetalliccorrosion,Typesofcorrosionandfactorsaffectingthe corrosionrate,Chemicalandelectrochemicaltheoryofcorrosion,Oxidefilmformationandits characteristics, tarnishingfogging and rusting, Prevention of corrosion byvarious methods.

(15 Lectures)

Unit:-V

POLYMERS:

1.Introductiontobasictermsusedinpolymerchemistryandtechnology.Monomers,Average degreeofpolymerisation,Averagemolecularweight.2.CharacteristicsofPolymersandtheir classification-Additionpolymersandtheirindustrialapplication–Polystyrene,PVC,PAN, Buna-S,Teflon.-Condensationpolymerandtheirindustrialapplication:Nylon6,Nylon6,6, Bakelite.

(8 Lectures)